

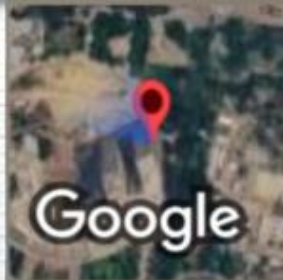
Introduction

What is a Data Model?

- A data model is a collection of concepts that describe:
- How data is created
- How data is organized
- How data is structured
- How users can access data

Importance

- Organizes information efficiently
- Reduces redundancy
- Improves data integrity
- Makes retrieval faster



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SIMATS ENGINEERING

Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases

CO1_AT3_Technical Presentation –CSA0502(DBMS)

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Introduction

- **What is a Data Model?**
- A data model is a collection of concepts that describe:
 - How data is stored
 - How data is organized
 - How data is connected
 - How users can access data
- **Importance**
 - Organizes information efficiently
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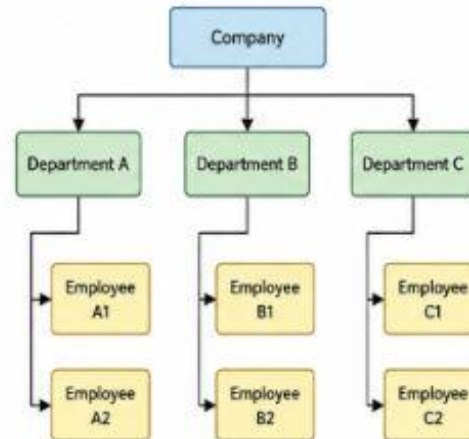
Types of Data Models

Three Major Data Models

- Hierarchical Data Model
- Network Data Model
- Relational Data Model

THREE MAJOR DATA MODELS

1. HIERARCHICAL DATA MODEL

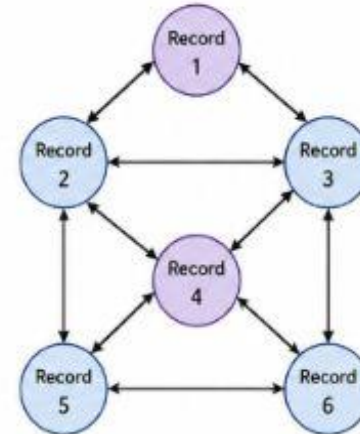


Characteristics:

- Tree-like structure
- One-to-many relationship
- Each child has only one parent

Example: Organization Hierarchy, File Systems

2. NETWORK DATA MODEL

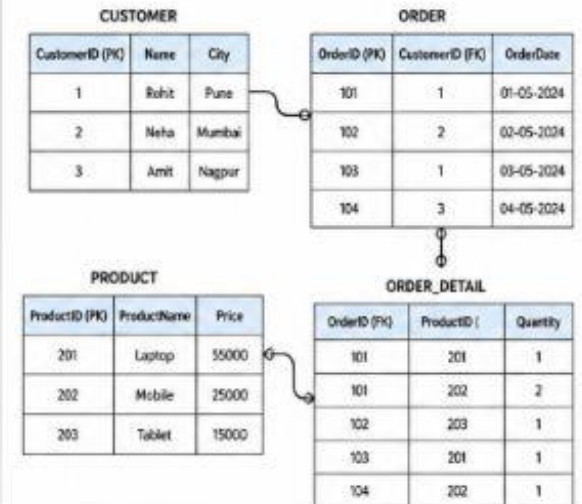


Characteristics:

- Graph-like structure
- Many-to-many relationship
- A record can have multiple parents and children

Example: Social Networks, Airline Routes, Telecommunications

3. RELATIONAL DATA MODEL



Characteristics:

- Table (relation) based structure
- Data represented in rows and columns
- Relationships using Primary Key (PK) and Foreign Key (FK)

Example: Banking Systems, Student Management Systems, E-commerce

Hierarchical Data Model

Definition

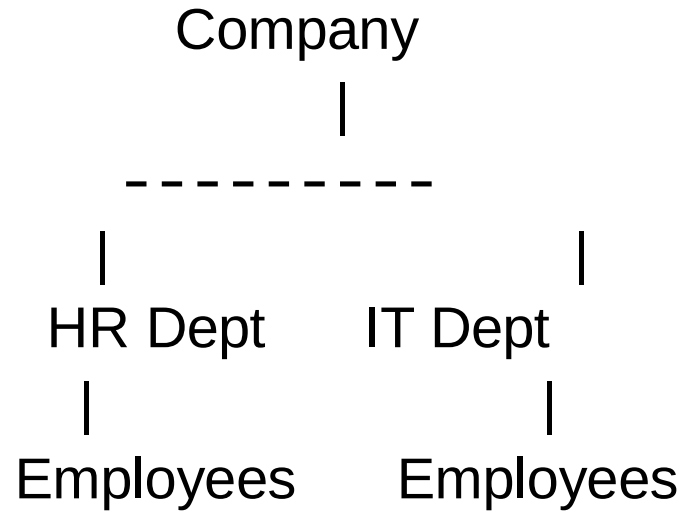
- The Hierarchical Data Model organizes data in a **tree-like structure**.
- Parent-child relationship
- One parent can have many children
- Each child has only one parent

Characteristics

- Tree structure
- One-to-many relationship
- Fast data retrieval
- Simple navigation

Hierarchical Data Model Diagram

Diagram Example



Explanation

- Company is the root.
- Departments are children.
- Employees belong to one department.

Advantages and Disadvantages of Hierarchical Model

Advantages

- ✓ Simple structure
- ✓ Fast searching
- ✓ High security
- ✓ Easy implementation

Disadvantages

- ✗ No many-to-many relationships
- ✗ Difficult to modify
- ✗ Data redundancy

Industry Use Cases of Hierarchical Model

Applications

- Organization Structure
- File Management System
- XML Documents
- Windows Registry
- Family Tree Systems

Network Data Model

Definition

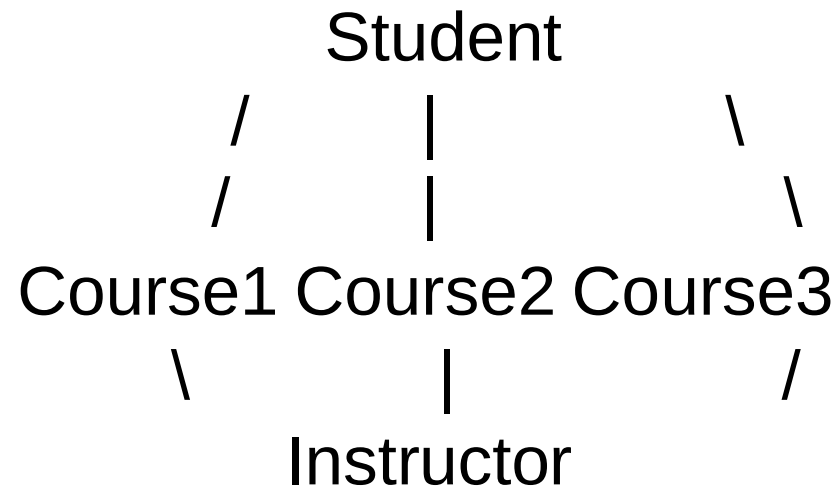
- The Network Model allows **many-to-many relationships**.
- Each record can have multiple parent records.

Features

- Graph structure
- Flexible relationships
- Multiple connections
- Efficient for complex databases

Network Data Model Diagram

Example



Explanation

- One student studies many courses.
- One instructor teaches multiple courses.

Industry Use Cases of Network Model

Applications

- Airline Reservation Systems
- Banking Networks
- Supply Chain Management
- Telecommunication Networks
- Transportation Systems

Relational Data Model

Definition

- The Relational Model stores data in **tables**.
- Each table contains:
 - Rows (Records)
 - Columns (Attributes)
- Tables are connected using **Primary Keys** and **Foreign Keys**.

Relational Model Diagram

Example

Student Table

Student_ID	Name
101	Rahul
102	Priya

Course Table

Student_ID	Course
101	DBMS
102	AI

Relationship:

Student_ID acts as Primary Key and Foreign Key.

Advantages and Disadvantages of Relational Model

Advantages

- ✓ Easy to understand
- ✓ Data consistency
- ✓ Reduced redundancy
- ✓ Powerful SQL queries
- ✓ High security

Disadvantages

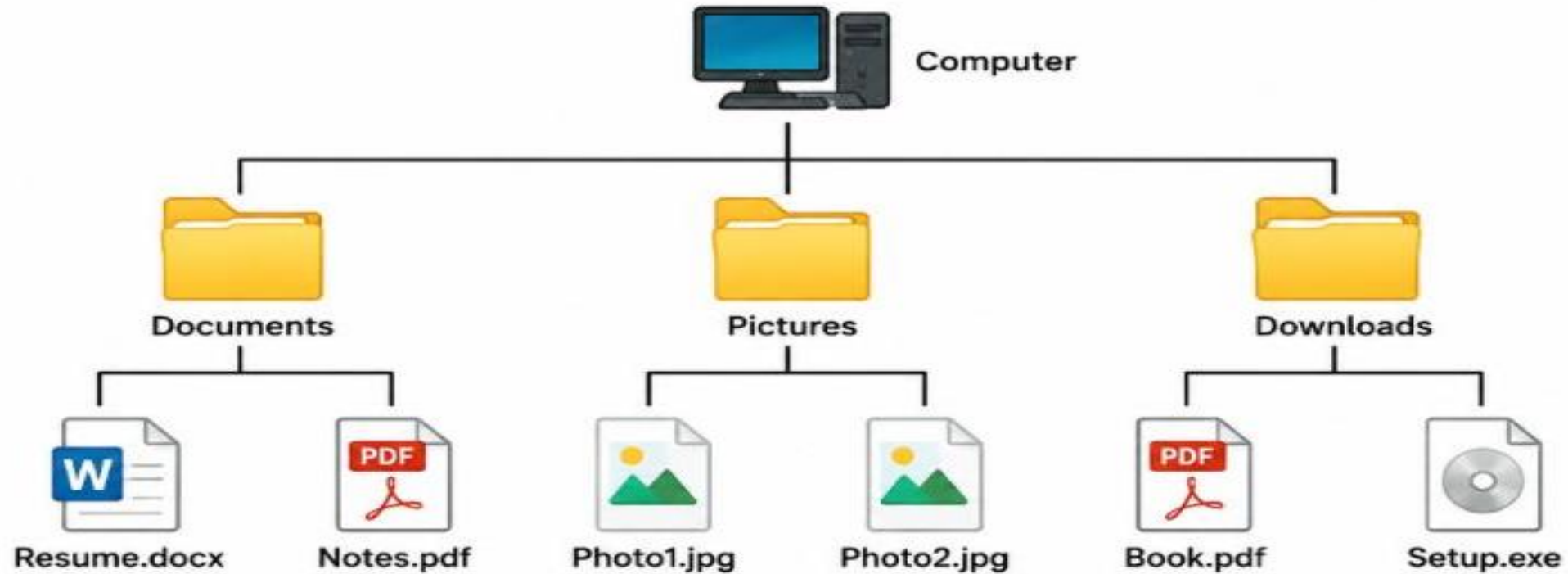
- ✗ Performance decreases with huge databases
- ✗ Complex joins
- ✗ Higher storage requirement

Comparison of Data Models:

Feature	Hierarchical	Network	Relational
Structure	Tree	Graph	Table
Relationship	One-to-Many	Many-to-Many	Keys
Complexity	Low	High	Medium
Flexibility	Low	High	High
Query Language	Limited	Complex	SQL

Real-World Comparison

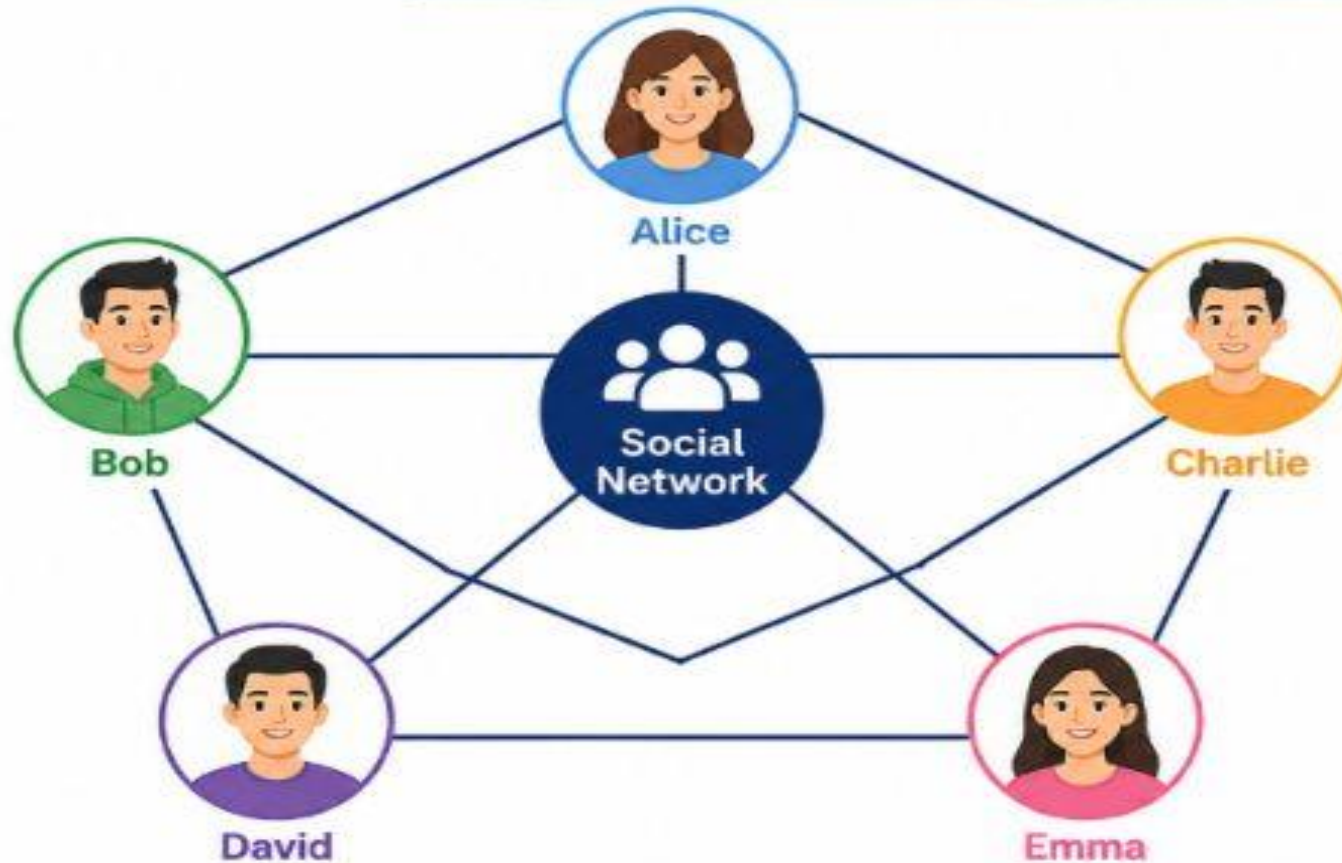
Hierarchical Data Model – Example



Computer is the root (parent). Folders are children of the root.
Files are children of folders. Each child has only one parent.

Network Data Model – Social Media Connections

Many-to-Many Relationships (Graph Structure)



What is it?

In a Network Data Model, each record (node) can have multiple parent and child records. It forms a graph structure that supports many-to-many relationships.



Example: Social Media

- Users can be friends with many other users.
- A user can have many friends.
- A friend can also have many other friends.

— Friend Connection



This model is ideal for systems where entities are interconnected in multiple ways, such as social networks, airline reservation systems, banking networks, etc.

Relational Data Model – Bank Database



Data is organized in tables (relations) and related using keys.



CUSTOMER			
<u>Customer_ID</u>	Name	Phone	Email
101	Rahul Sharma	9876543210	rahul@email.com
102	Priya Singh	8765432109	priya@email.com
103	Amit Verma	9123456780	amit@email.com

ACCOUNT			
<u>Account_No</u>	Customer_ID	Account_Type	Balance
5001	101	Savings	75,000.00
5002	101	Current	1,20,000.00
5003	102	Savings	50,000.00
5004	103	Current	80,000.00

CUSTOMER.Customer_ID (PK) → ACCOUNT.Customer_ID (FK)

LOAN			
<u>Loan_ID</u>	Customer_ID	Loan_Type	Amount
L101	101	Home Loan	15,00,000
L102	102	Car Loan	5,00,000
L103	103	Personal Loan	2,00,000

CUSTOMER.Customer_ID (PK) → LOAN.Customer_ID (FK)

TRANSACTION				
<u>Trans_ID</u>	Account_No	Trans_Date	Amount	Type
T1001	5001	01-05-2024	10,000.00	Credit
T1002	5001	03-05-2024	2,000.00	Debit
T1003	5002	04-05-2024	15,000.00	Credit
T1004	5003	05-05-2024	5,000.00	Debit
T1005	5004	06-05-2024	20,000.00	Credit

ACCOUNT.Account_No (PK) → TRANSACTION.Account_No (FK)

Key Points

- ✓ Data is stored in tables (rows and columns).
- ✓ Each table has a Primary Key.
- ✓ Tables are related using Foreign Keys.
- ✓ Reduces redundancy and improves data integrity.

How it works in a Bank



A customer can have multiple accounts and loans.



An account can have multiple transactions.



This structure minimizes data redundancy and ensures data consistency.

Future of Data Models

Modern databases combine traditional models with new technologies.

Examples:

- Cloud Databases
- NoSQL Databases
- Graph Databases
- Distributed Databases

Benefits:

- Scalability
- High availability
- Big Data support

Conclusion

- Different data models provide different ways to organize and manage data. The choice of a data model depends on the application's complexity, relationship requirements, and performance needs. While hierarchical and network models are useful for specific applications, the relational model remains the most popular due to its simplicity, flexibility, and powerful SQL support.

Thank you